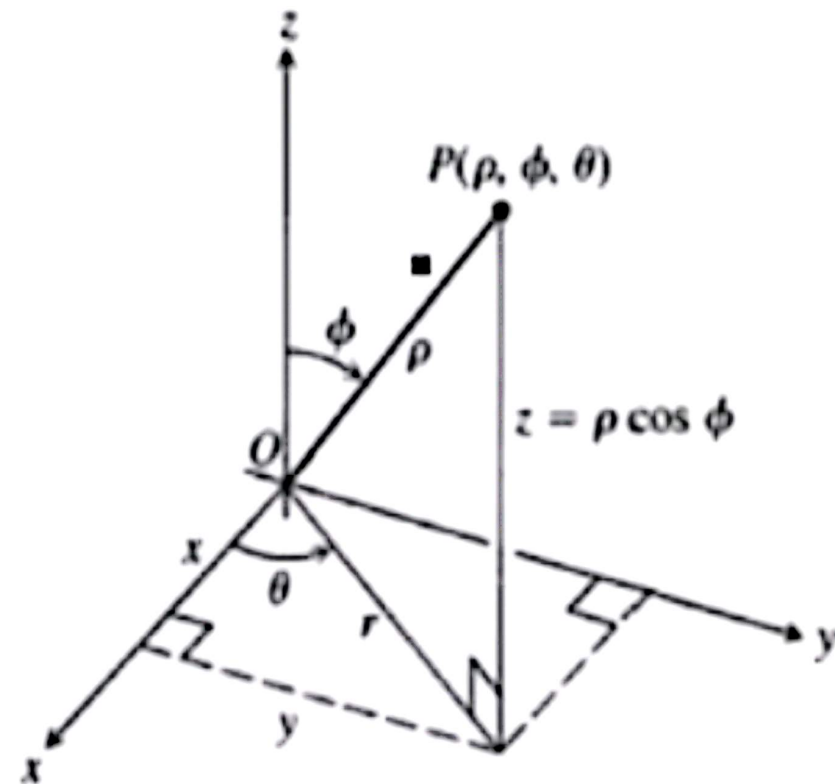


Spherical Coordinates

Spherical Coordinates locate points in space with angles and a distance by ordered triples (ρ, ϕ, θ) in which

- 1) ρ is the distance from P to the origin, ($\rho \geq 0$)
- 2) ϕ is the angle $\vec{OP}$ makes with the positive z -axis ($0 \leq \phi \leq \pi$)
- 3) θ is the angle from cylindrical coordinates.



$$\begin{aligned}
 x^2 + y^2 &= \rho^2 \sin^2 \phi \cos^2 \theta + \rho^2 \sin^2 \phi \sin^2 \theta \\
 &= \rho^2 \sin^2 \phi (\cos^2 \theta + \sin^2 \theta) = \rho^2 \sin^2 \phi \\
 x^2 + y^2 + z^2 &= \rho^2 \sin^2 \phi + \rho^2 \cos^2 \phi = \rho^2.
 \end{aligned}$$

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$r = \rho \cos(\pi/2 - \phi) = \rho \sin \phi$$

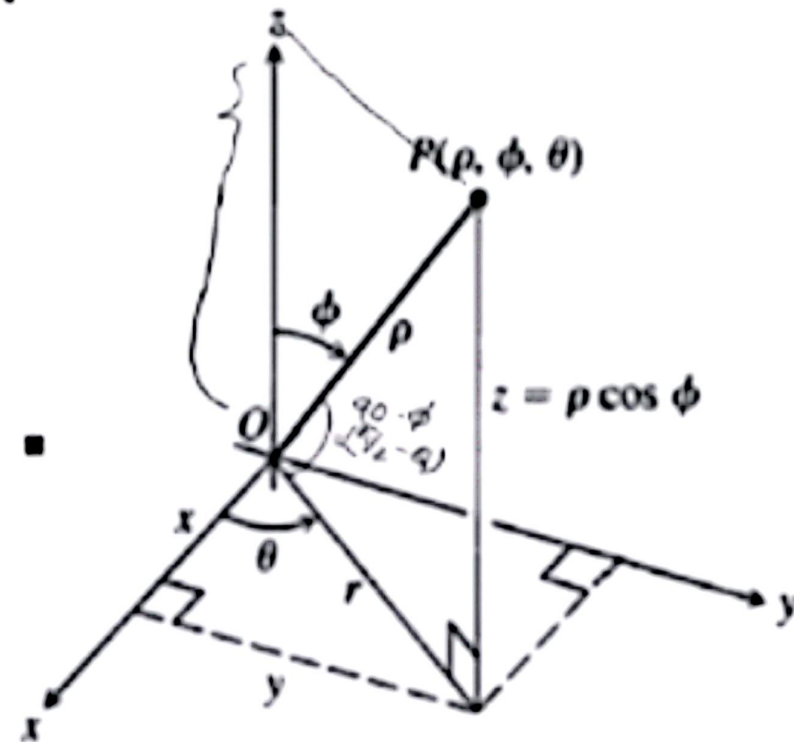
$$z = \rho \sin(\pi/2 - \phi) = \rho \cos \phi$$

$$x = \rho \sin \phi \cos \theta$$

$$y = \rho \sin \phi \sin \theta$$

$$\rho = \sqrt{x^2 + y^2 + z^2}$$

$$\tan \theta = y/x$$



Equations Relating Spherical Coordinates to Cartesian and Cylindrical Coordinates

$$r = \rho \sin \phi, \quad x = r \cos \theta = \rho \sin \phi \cos \theta,$$

$$z = \rho \cos \phi, \quad y = r \sin \theta = \rho \sin \phi \sin \theta,$$

$$\rho = \sqrt{x^2 + y^2 + z^2} = \sqrt{r^2 + z^2}$$

EXAMPLE 6 Find a spherical coordinate equation for the sphere

$$x^2 + y^2 + (z - 1)^2 = 1.$$

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$$x^2 + y^2 + z^2 - 2z + 1 = 1$$

$$x^2 + y^2 + z^2 - 2z = 0$$

$$x^2 + y^2 + z^2 = 2z$$

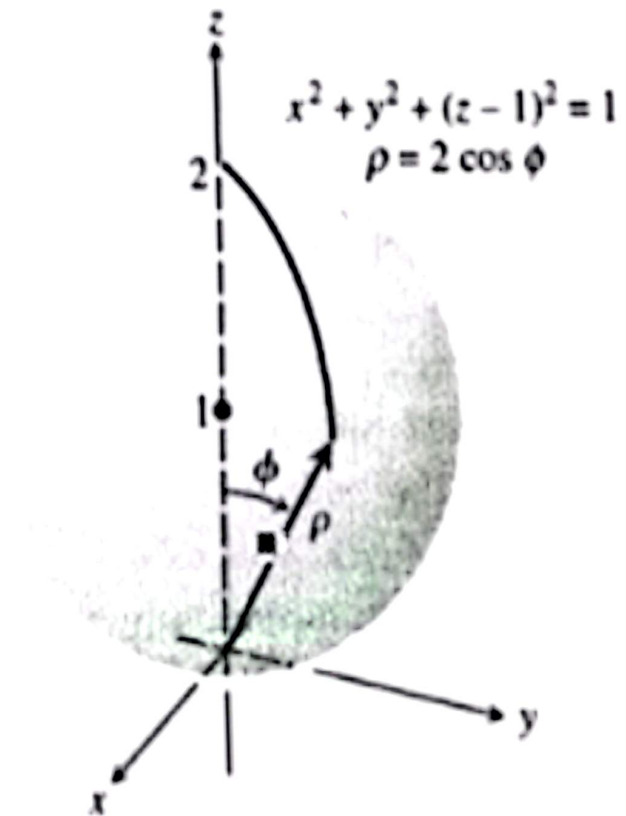
$$\rho^2 = 2\rho \cos \phi$$

$$\rho(\rho - 2 \cos \phi) = 0$$

$$\rho = 2 \cos \phi$$

EXAMPLE 6 Find a spherical coordinate equation for the sphere

$$x^2 + y^2 + (z - 1)^2 = 1.$$



EXAMPLE 7 Find a spherical coordinate equation for the cone $z = \sqrt{x^2 + y^2}$

$$z = \sqrt{x^2 + y^2}$$

$$z^2 = x^2 + y^2$$

$$z = \rho \cos \phi, \quad x = \rho \sin \phi \cos \theta, \\ y = \rho \sin \phi \sin \theta.$$

$$\begin{aligned} \rho^2 \cos^2 \phi &= \rho^2 \sin^2 \phi \cos^2 \theta + \rho^2 \sin^2 \phi \sin^2 \theta \\ &= \rho^2 \sin^2 \phi (\cos^2 \theta + \sin^2 \theta) \\ &= \rho^2 \sin^2 \phi \end{aligned}$$

$$\frac{\rho^2 \sin^2 \phi}{\rho^2 \cos^2 \phi} = 1 \quad \Rightarrow \quad \frac{\sin^2 \phi}{\cos^2 \phi} = 1 \quad \phi = \frac{\pi}{4}$$

EXAMPLE 7 Find a spherical coordinate equation for the cone $z = \sqrt{x^2 + y^2}$

